If we view the map $\mathscr{F}_t$ as a change of variables, we have

$$\mathrm{Vol}(\mathcal{S}(t)) = \int_{\mathcal{S}} |D| \mathrm{d}\omega,$$

where $D = \det\left(\frac{\partial \mathscr{F}_t}{\partial z}\right)$.

To understand where Liouville's theorem comes from, recall that the variational equations of the last chapter are a system of ordinary differential equations for $\boldsymbol{W}(t) = \mathscr{F}_t'(z(t))$:

$$\frac{\mathrm{d}\boldsymbol{W}}{\mathrm{d}t} = \boldsymbol{f}'(z(t))\boldsymbol{W}.$$

Thus

$$\dot{\boldsymbol{W}}\boldsymbol{W}^{-1} = \boldsymbol{f}'(z(t)).$$

Now let $D = \det(\boldsymbol{W})$. One can show (see Exercise 5) that

$$\frac{\dot{D}}{D} = \mathrm{tr}\left(\dot{\boldsymbol{W}}\boldsymbol{W}^{-1}\right).$$

This implies that

$$\dot{D} = \mathrm{div}(\boldsymbol{f}(z(t)))D,$$

and thus

$$D(t) = D(0)e^{\int_0^t \mathrm{div}(\boldsymbol{f}(z(s)))\mathrm{d}s}.$$

In particular, if $\mathrm{div}\boldsymbol{f} \equiv 0$, we see that $D \equiv D(0) = 1$ and it follows that the volume is constant.

Liouville's theorem may be summarized compactly as:

$$\nabla \cdot \boldsymbol{f} = 0 \Rightarrow \det \mathscr{F}_t' = 1.$$

Example 2.3 A 1-d oscillator with Lennard-Jones potential is described by the equations

$$\begin{aligned} \dot{q} &= p, \\ \dot{p} &= -\varphi_{\mathrm{LJ}}'(q). \end{aligned}$$

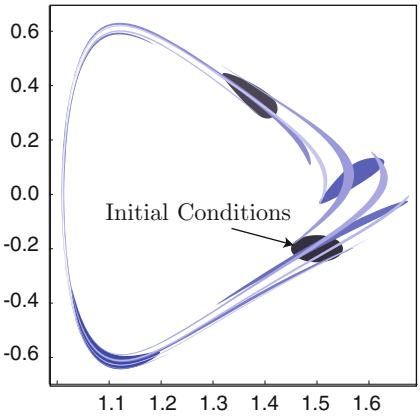


Fig. 2.6 Filamentation of an evolving disk under the flow map of the Lennard-Jones oscillator. Lighter regions represent later images. Despite the increasing complexity of the shapes, each snapshot has the same area

As a consequence of energy conservation, any bounded individual trajectory of this system will be a periodic orbit. Consider the propagation of a small disk of initial conditions,

$$\mathcal{R} = \{(q_0, p_0) | \|(q_0, p_0) - (1.5, -0.2)\| \leq 0.5\}.$$

In Fig. 2.6, we have plotted the sets $\mathscr{F}_t(\mathcal{R})$, for $t = 0, 1, 2, \ldots, 10$, using lighter tones to represent later snapshots. The sets become extremely elongated along the orbital direction.

At all times the energies stay bounded within the interval

$$-0.0960 \leq H(q, p) \leq -0.0669,$$

which is defined by the range of energies in the initial conditions.

Even as the sets become more and more elongated, they always maintain the same total area. This is the direct consequence of Liouville's theorem.

2.3.2 *Volume Preserving Numerical Methods*

The next question is what happens to the volume of a set of points in phase space, which would be conserved by the dynamical system, when we use a numerical method to approximate its evolution.

Consider a linear differential equation system in $\mathbb{R}^m$,

$$\dot{z} = \boldsymbol{S}z,$$

for some matrix $\boldsymbol{S} \in \mathbb{R}^{m \times m}$. The condition for the flow of this system to conserve volume is just that the trace of $\boldsymbol{S}$ (which is the divergence of the vector field